

Session 24: Analyze & Synthesize

You've planned. You've researched. You've explored. Today is the most demanding session of your capstone: you look at AI in your domain through **5 lenses**, then draft the two new sections - **Bias & Fairness** and **Safety & Governance** - together with your mentor.



Objectives

By the end of this session, you'll have accomplished four critical milestones:

01

Understood the 5 Lenses

For analyzing AI in your domain

02

Drafted Bias & Fairness

With mentor coaching 150–200 words

03

Drafted Safety & Governance

With mentor coaching 150–200 words

04

Mapped Homework

Clear plan for everything coming next

- ① In-class focus today: the two new analysis sections. Everything else polish, Conclusion, Abstract, AI Attribution, full Gamma deck - happens in homework. Mentor scaffolds both section drafts heavily. This is the heaviest writing day of the capstone.

Today's Work:

In Class vs. Homework

In Class Today

-  Bias & Fairness section
-  Safety & Governance section

Homework

-  Polish all 6 prior sections
-  Draft Conclusion & Future Work
-  Write Final Abstract
-  Write AI Attribution Statement
-  Build all 9 Gamma slides
-  Practice presentation aloud

Homework Review - What Mentor Will Flag

Your mentor has read your S23 homework. Specific feedback is coming on these key areas:



Generated-Sounding Phrasing

Sentences that sound polished but aren't in **YOUR** voice



Vague Claims

Places that need specifics + evidence to back them up



Surprises ↔ Analysis

Your 3 surprises will be connected to bias/safety today



Strongest Observations

What to anchor today's analysis around

The 5 Lenses - Your Analysis Toolkit

You'll look at AI in your domain through 5 different angles. Two become full sections; three get woven in.

Lens	What It Looks At	How It Appears
 Bias & Fairness	Who AI works for · who it fails	Full section (drafted today)
 Safety & Governance	What harms · who's responsible	Full section (drafted today)
 Ethics	What values are at stake · who benefits · who could be hurt	Woven in
 Human-in-the-Loop	Where humans oversee · where they should	Woven in
 Voice Integrity	Whether YOUR paper sounds like YOU	Across entire paper

Lens 1 - Bias & Fairness

You learned bias in Session 20. Today you apply it to AI in **YOUR** domain. Remember: you're analyzing how AI is **USED** in your domain right now - not a model you built.



Data Bias

What's **IN** and **MISSING** from the training data



Algorithm Bias

How the AI processes information **unfairly**



Human Bias

Assumptions the **people who built it** brought in

Bias Questions to Ask About YOUR Domain

Work through these 4 questions to build your Bias section. Your sources + your exploration + your testers = your Bias evidence.

1 Training Populations

Who is AI in your domain trained on? Who's overrepresented?
Who's **MISSING**?

3 Hands-On Patterns

What did YOUR exploration reveal about who the tool works for vs. doesn't?

2 Published Failures

Where do your sources show AI in your domain failing? Who does it consistently underserve?

4 User Testing Signals

What did your testers notice about who the tool worked for or not?

Bias - A Concrete Example

Domain: Food Allergy Management for Children

What Sources Revealed

Allergen-detection AI is mostly trained on **Western foods**. This means it underperforms on non-Western cuisines.

What Exploration Confirmed

The classifier handled Snickers and peanut butter reliably - but had **NO training data** for Asian peanut sauces, African peanut stews, or Middle Eastern peanut-based sweets.

The Specific Claim

"This data gap means the tool could fail for the families whose foods include these dishes - families who are NOT less in need of allergen support. Just less represented in training data."

Section Draft 1 - Draft Your Bias & Fairness Section!

Together with your mentor, draft your Bias & Fairness section using this 4-part structure:



State the Issue

Name the most important bias issue in YOUR domain



Exploration Evidence

What your hands-on revealed about bias patterns



Source Evidence

What your sources say about who AI works for/against



Connect to Frameworks

Link to broader bias ideas where relevant



Target

150–200 words



Support

Mentor coaches throughout

Lens 2 - Safety & Governance

You learned safety in Session 21. Today you apply it to YOUR domain. These are **real risks** - don't sugar-coat them.

Misuse

How could AI in this domain be used to harm people?

Privacy

What personal data is collected? How could it leak?

Accountability

Who's responsible when something goes wrong?

Safety Questions to Ask About YOUR Domain

Work through these 4 questions for your Safety section. Same pattern: literature evidence + exploration evidence + user testing observations.

→ Realistic Misuse

How could AI in your domain be misused, intentionally or not?

→ Sensitive Data

What personal info do these AI apps collect? How could it be exposed?

→ Accountability Gap

When something goes wrong, who's responsible? Is there a clear answer?

→ User Concerns

What safety risks did YOUR exploration or testers surface?

Safety - A Concrete Example

Domain: Food Allergy Management for Children

⚠️ Most Serious Concern: Misplaced Trust

Sources reported that consumer-facing allergen detection often shows **high confidence even when WRONG**.

🔧 What Exploration Revealed

When the classifier misidentified a Snickers bar as peanut-free, it did so with **HIGH confidence**. A parent or child trusting that = a life-threatening reaction.

⚖️ The Accountability Gap

If a child consumes an allergen, who's responsible?

- The parent?
- The app maker?
- The platform?

There's **no clear rule yet**.

Section Draft 2 - Draft Your Safety & Governance Section!

Use this 4-part structure with your mentor. Weave in Ethics + Human-in-the-Loop here - they don't get separate sections.



Name the Concerns

Most serious safety risks in your domain



Connect to YOUR Work

Your hands-on + testing observations



Address ALL Three Categories

Misuse, privacy, accountability - even briefly



Note Human Oversight

Where humans need to stay in charge



Target

150–200 words



Weave In

Ethics + Human-in-the-Loop

Lenses 3, 4, 5 - How to Weave Them In

Three lenses don't get their own sections - they integrate into Bias and Safety writing.

Ethics

Add 1–2 sentences about who benefits, who could be harmed (especially those who can't say no - kids, patients, elderly), and what values are in tension. Place **inside Bias OR Safety**.

Human-in-the-Loop

Note where human oversight exists, where it's missing, where it **SHOULD** be added. Place **inside Safety**.

Voice Integrity

The lens you apply to YOUR paper - every sentence must sound like YOU. Read aloud. If it doesn't sound like you → rewrite. Applies **across your entire paper**.

AI-Sounding Phrases to Watch For

Generated phrasing follows predictable patterns. Learn to spot them - then rewrite, don't just swap one phrase for another.

Overly Polished

Doesn't match your earlier writing style or voice

Hollow Phrases

"In today's rapidly evolving landscape" or "leveraging cutting-edge technology"

Unusual Vocabulary

Words you wouldn't say in conversation

Generic Claims

Could appear in ANY paper on ANY topic

⚠ When you spot them: **REWRITE**. Don't swap one phrase for another. Actually think through what you mean. 📌
You'll be checking for these during homework polish.

Walk-Through - What Homework Looks Like

01

Polish All 6 Prior Sections

Read aloud → Check every claim has a citation OR comes from your exploration → Check transitions → Cut filler.

03

Write Final Abstract

~80 words. Replace S22 placeholder.

5 components: Problem · Question · Method · Findings · Implications.

05

Build All 9 Gamma Slides

Full presentation deck built in Gamma.

02

Draft Conclusion & Future Work

100–150 words. Restate investigation · Summarize findings · State takeaway · Point forward. No new findings.

04

Write AI Attribution Statement

Disclose what AI tools helped and what YOU did yourself.

06

Practice Presentation Aloud

Run through at least twice.

Polish - How to Do It Right

When you polish at home, work in this exact order. Don't skip steps - each one catches different problems.

Read Aloud for Voice

Does it sound like **YOU**?

Check Every Claim

Does it have a citation OR come from your exploration?

Check Transitions

Does each section flow into the next?

Cut Filler

Remove words that don't earn their place

⚠️ Polish until it's clearly yours, well-cited, flows. Then **STOP**. Don't polish until it's "perfect." That's a trap. Voice and clarity matter more than fancy.

Drafting the Conclusion

Your Conclusion summarizes - it doesn't introduce new findings. Everything in here should connect back to something you already showed.

Restate the Investigation

What you investigated and why it mattered

Summarize Findings

What sources + exploration revealed

State the Takeaway

Most important thing you learned

Point Forward

What comes next - this is your Future Work

Target

100–150 words

Rule

No new findings

Writing Your FINAL Abstract

Now you replace the S22 placeholder with the REAL Abstract. Use this 5-component formula.

Target: ~80 words. 📌 Last thing you write - but appears **FIRST** in your paper.

Component	Now You Can Write
Problem	Why your domain mattered
Question	Your sharpened research question
Method	How you investigated
Findings	What you ACTUALLY found
Implications	What it means for your domain

❗ The Abstract is the last thing you write - but it appears **first** in your paper. Write it after everything else is done.

AI Attribution Statement

A short statement disclosing what AI tools helped - and what YOU did yourself. It's not a confession. It's **TRANSPARENCY** - the standard for modern research.

For Each AI Tool You Used:

- Which tool
- What you used it for
- How you verified its output

Also State What YOU Did:

- Drafted all sections
- Polished in your voice
- Did the hands-on exploration
- Conducted user testing
- Decided what claims to make

AI Attribution - Example

Here's what a complete AI Attribution Statement looks like. Yours should reflect what YOU actually used.

"AI tools assisted in this investigation. Perplexity helped discover candidate sources. NotebookLM was used to verify content and extract findings, with all extractions cross-checked. Gemini compared candidate sources during filtering. Teachable Machine was the subject of the hands-on exploration. ChatGPT was used briefly for brainstorming on framing, not for content generation. All paper sections were drafted, polished, and revised by me. Where AI tools suggested phrasing, I rewrote in my own voice. My mentor provided guidance and feedback throughout but did not write any sections of this paper."

📌 Yours should reflect what **YOU** actually used - be specific and honest about each tool's role.

Build Your Gamma Presentation

You'll build all 9 slides in homework. Use Gamma - you saw it in Session 17. ⌚ Build all 9 in homework.

#	Slide	Content
1	Title	Paper title + name + grade
2	Hook	Most striking finding
3	Research Question	Your sharpened question
4	Background	3 key findings from Related Work
5	Methodology	How you investigated
6	Results	Your synthesis
7	Bias & Safety	Main concerns
8	Conclusion	Main takeaway + future work
9	Thank You	Closing

Practice Your Presentation

After your 9 slides are built - practice. Run through your presentation aloud at least twice before Session 25.



Read Each Slide

Speak naturally - don't read robotically



Time Yourself

Aim for 10–15 minutes total



Listen Critically

Does it sound like YOU? Are you explaining or just reading?



Rewrite Stumbles

Spots where you stumble → rewrite those slides

 Practice at least **twice** before Session 25.

Outputs Checklist - End of Today

By End of This Session

- ☒ Bias & Fairness section drafted (150–200 words, mentor-coached)
- ☒ Safety & Governance section drafted (150–200 words, Ethics + HITL woven in)
- ☒ Clear understanding of the 5 Lenses
- ☒ Clear plan for homework

Coming in Homework

- ☐ Polish all 6 prior sections
- ☐ Draft Conclusion (100–150 words)
- ☐ Write Final Abstract (80 words)
- ☐ Write AI Attribution Statement
- ☐ Build all 9 Gamma slides
- ☐ Practice presentation aloud

Next Time - Session 25: Deliver & Reflect 🎓

This is the final session of the entire course. Come ready.

Present

Your full 10–15 minute investigation presentation

Defend

Mentor asks journal-reviewer-style questions

Reflect

On the 25-session journey - what you learned, how you grew

Receive

Your portfolio - paper + presentation + reflection